

POLTAVA, Ukraine

WMO: 335060

Lat: 49.6094N

Lon: 34.5447E

Elev: 160

StdP: 99.42

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-15.8	-22.1	0.5	-17.9	-19.3	0.7	-15.4	10.2	-8.7	9.2	-6.7	2.7	50	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	32.0	20.1	30.2	19.6	28.6	19.0	21.7	28.8	20.9	27.8	20.1	26.5	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.4	24.7	18.6	13.7	23.9	17.7	13.0	23.1	64.1	28.8	61.1	27.8	58.3	26.5	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.6	6.4	DB	-21.8	34.3	3.6	2.0	-24.5	35.8	-26.6	36.9	-28.6	38.1	-31.3	39.5	(4)
(5)			WB	-22.1	23.2	3.6	1.1	-24.7	24.0	-26.8	24.7	-28.8	25.3	-31.4	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.7	-3.5	1.7	10.0	16.1	19.6	21.7	20.9	15.2	8.4	2.4	-2.5	(6)	
	DBStd	10.52	5.72	5.82	4.55	4.32	4.37	3.57	3.06	3.64	4.16	4.60	4.80	5.46	(7)	
	HDD10.0	1856	454	377	257	51	6	0	0	0	6	84	232	389	(8)	
	HDD18.3	3773	713	611	514	249	96	28	6	14	109	307	479	647	(9)	
	CDD10.0	1442	0	0	1	53	195	288	363	339	164	36	3	0	(10)	
	CDD18.3	316	0	0	0	1	27	66	110	96	17	1	0	0	(11)	
	CDH23.3	2727	0	0	0	11	239	524	924	881	142	5	0	0	(12)	
	CDH26.7	858	0	0	0	1	53	137	308	329	28	1	0	0	(13)	
(14) Wind	WSAvg	2.9	3.2	3.3	3.5	3.1	2.7	2.7	2.4	2.5	2.5	2.6	3.0	3.1	(14)	
(15) Precipitation	PrecAvg	566	44	34	37	40	52	59	66	44	47	47	47	46	(15)	
	PrecMax	863	136	79	96	99	110	204	222	103	166	198	119	126	(16)	
	PrecMin	337	6	1	5	11	12	6	14	5	2	0	6	3	(17)	
	PrecStd	127	28	18	21	23	27	37	45	26	36	42	28	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	8.5	16.0	25.0	30.6	31.8	34.0	35.8	29.6	23.0	15.3	9.5	(19)	
		MCWB	5.8	6.3	9.0	14.1	18.5	19.8	21.0	20.2	17.4	15.3	12.0	8.8	(20)	
	2%	DB	4.6	6.1	12.6	21.5	28.1	29.8	31.9	32.6	27.1	20.1	12.5	7.2	(21)	
		MCWB	3.9	4.9	7.4	12.4	16.9	19.7	20.7	19.8	16.3	14.1	10.5	6.5	(22)	
	5%	DB	3.2	4.3	10.2	19.4	26.1	28.2	30.0	30.2	24.7	17.1	10.5	5.6	(23)	
		MCWB	2.4	3.2	6.2	11.5	16.7	19.3	20.1	19.3	15.6	12.9	9.2	5.0	(24)	
	10%	DB	1.9	2.9	8.0	17.2	23.9	26.5	28.2	28.0	22.3	14.9	8.6	3.9	(25)	
		MCWB	1.3	2.0	5.0	10.5	15.7	18.5	19.7	18.5	15.0	11.8	7.4	3.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.0	9.9	15.2	20.2	22.2	22.8	22.6	19.2	16.8	12.6	8.9	(27)	
		MCDB	6.4	8.0	14.6	23.2	28.4	28.1	30.3	29.9	25.7	21.1	14.5	9.4	(28)	
	2%	WB	4.1	5.1	8.1	13.4	18.6	21.1	21.9	21.2	17.7	14.8	10.9	6.6	(29)	
		MCDB	4.5	6.1	11.5	19.4	25.3	27.6	29.1	28.9	24.0	18.7	12.4	7.0	(30)	
	5%	WB	2.5	3.3	6.6	12.3	17.5	20.2	21.1	20.2	16.8	13.4	9.3	5.0	(31)	
		MCDB	3.1	4.0	9.4	17.3	23.5	26.2	28.0	27.7	22.6	16.3	10.3	5.5	(32)	
	10%	WB	1.4	2.1	5.3	11.4	16.4	19.2	20.4	19.3	15.7	12.1	7.6	3.4	(33)	
		MCDB	1.8	2.8	7.7	15.9	22.3	25.0	26.6	26.4	20.4	14.5	8.5	3.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.1	6.1	8.7	9.8	9.2	9.7	10.3	9.1	7.1	4.5	4.1	(35)	
		MCDBR	3.8	5.6	9.8	11.7	12.6	11.4	12.0	13.1	12.7	10.6	6.7	4.6	(36)	
	5% WB	MCWBR	3.5	4.7	6.0	5.6	5.1	4.9	4.4	4.4	5.4	5.6	5.0	4.3	(37)	
		MCWBR	3.7	5.2	8.7	9.9	11.0	10.1	10.5	11.3	10.6	8.9	6.0	4.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.318	0.347	0.378	0.446	0.410	0.401	0.431	0.429	0.401	0.365	0.348	0.321	(40)	
		taud	2.183	2.129	2.184	2.086	2.271	2.316	2.239	2.233	2.279	2.356	2.301	2.262	(41)	
	Edn at Noon	Edn at Noon	698	769	817	797	849	859	824	806	787	753	665	643	(42)	
		Edh at Noon	78	106	120	148	128	123	131	126	109	85	71	64	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.94	1.80	3.01	4.35	5.75	6.17	6.01	5.36	3.65	2.11	0.90	0.61	(44)	
	RadStd	RadStd	0.18	0.31	0.42	0.41	0.49	0.54	0.38	0.40	0.46	0.32	0.17	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.76	N/A	N/A	+0.91	N/A	N/A	-157	-207	+110	+68	(46)
(47)	Regional (9 neighbors)	+0.77	N/A	N/A	+1.04	N/A	N/A	-147	-200	+122	+78	(47)

Nomenclature: See separate page